

Rising Waters: A Machine Learning Approach to Flood Prediction

Scalability & Future Plan

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited historical dataset	Reduced generalization	High
2	Local deployment only	Limited accessibility	Medium
3	Single-region prediction	Reduced scalability	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local users	Cloud deployment on IBM Cloud
2	Data Storage	Local files	Cloud database integration
3	Performance	Single server	Distributed cloud infrastructure
4	Security	Basic authentication	Advanced access control and encryption

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Real-time weather API integration	3 Months	Improved prediction accuracy
Phase 2	Mobile Application Development	6 Months	Wider accessibility

Phase 3	Multi-region Flood Monitoring	12 Months	Large-scale deployment
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